

NASSIM LATTAB

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EDUCATION

Université Paris Cité

Paris, France

MSc in Computer Science — Artificial Intelligence

Sep. 2023 – Sep. 2025

- Graduated **1st out of 16** — with Honors
- Coursework: Deep Learning, Reinforcement Learning, NLP (LLMs, Transformers), Data Science, MLOps
- Includes courses taught by professors from Sorbonne Université and CentraleSupélec

Université Paris Cité

Paris, France

BSc in Computer Science

Sep. 2020 – May 2023

- Graduated **6th out of 117** — with Honors

EXPERIENCE

AI Research Intern — Deep Learning & Computer Vision

Feb. 2025 – Aug. 2025

Sodern (ArianeGroup)

Limeil-Brévannes, France

- Architected and trained a U-Net model for semantic segmentation of artifacts on astronomical images
- Developed a U-Net-based inpainting pipeline to reconstruct corrupted regions in embedded space systems
- Optimized model performance using Bayesian hyperparameter search with Optuna
- Implemented post-training quantization, reducing model size by 50% for lightweight embedded deployment
- Engineered a TFLite and ONNX inference pipeline for onboard deployment in resource-constrained environments
- Leveraged SHAP (SHapley Additive exPlanations) for model explainability and feature importance analysis

PROJECTS

Image Generation with Diffusion Models | *PyTorch, DDPM, DDIM, Computer Vision*

2026

- From-scratch implementation of DDPM and DDIM samplers in PyTorch, custom U-Net with 14M parameters
- FID/IS evaluation on 50k samples (clean-fid, torchmetrics), FID 10.74 (DDPM-1000) on FashionMNIST
- DDIM-50 with 20× faster inference (FID 19.39), EMA-smoothed weights for sampling

ETTh1 Time Series Forecasting (LSTM + RevIN) | *PyTorch, RNN, Forecasting*

2026

- Implemented a multi-horizon LSTM on ETTh1 (96/192/336/720), achieving 1.63°C MAE at horizon 96
- Integrated RevIN (ICLR 2022) to address distribution shift, reducing MSE by a factor of 3.5
- Analyzed performance degradation across forecasting horizons: MAE 1.63°C (96h) → 2.27°C (720h)

IMDb Sentiment Classification (RoBERTa) | *PyTorch, HuggingFace Transformers, NLP*

2026

- Fine-tuned RoBERTa-base (125M parameters) on IMDb, achieving 94.12% accuracy on the 25k-review test set
- Integrated RoBERTa BPE tokenization with dynamic batching for GPU training and inference
- Analyzed IMDb dataset limitations leading to highly polarized predictions due to the absence of a neutral class

CIFAR-10 Image Classification (ResNet18) | *PyTorch, CNN, Computer Vision*

2026

- Designed a ResNet18 from scratch (skip connections, BasicBlocks) reaching 95% accuracy on CIFAR-10
- Comparative study: SimpleCNN baseline (1M+ parameters, 88%) vs ResNet18 (11M+ parameters, 95%)
- Analyzed the impact of skip connections on training stability and network depth

TECHNICAL SKILLS

Languages: Python, C++, Bash, SQL

ML / AI: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Transformers, Data Science

Frameworks: PyTorch, TensorFlow, HuggingFace, Scikit-learn, Optuna, TensorBoard, FastAPI, Flask

Libraries: NumPy, Pandas, OpenCV, Matplotlib, NLTK, SHAP

DevOps & Tools: Docker, Kubernetes, Git, Linux, CUDA, ONNX, TFLite

LANGUAGES

French: Native | **English:** Professional working proficiency